

Data Science

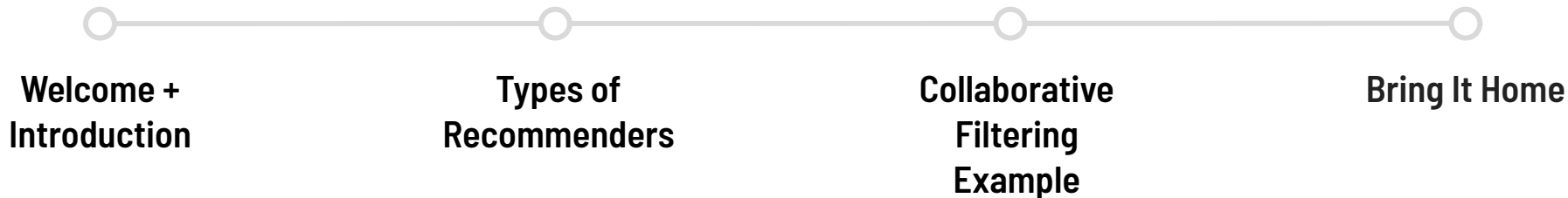
Recommender Systems

Advanced Modeling



GENERAL ASSEMBLY

Lesson roadmap



Learning objectives



1

Explain the difference between collaborative filtering and content-based systems.

2

Understand Cosine Similarity.

4

Explain the math behind recommendation engines.

3

Create an item-based movie recommender.



Trivia



What % of watch time on YouTube comes from recommendations...?

Types of Recommenders

What is a recommender system?

Simply put, a **recommender system** is a system that can help recommend new items in a service to someone. You probably interact with many of them per day:

- ❑ Amazon: “You might also like...”
- ❑ Netflix: “Recommended for Tim...”
- ❑ Goodreads: “Because you liked X, you might like Y.”
- ❑ Spotify: “Tim’s Daily Mix”



The Netflix Prize

In 2006, little-known DVD rental service **Netflix** issued a challenge: to beat their recommender system and win \$1,000,000! A winner was declared in 2009. Netflix ended up not using their algorithm in the end.

You can read more about it [here](#)!



Different Kinds of Recommenders

1

Collaborative
filtering

2

Content-based

3

Knowledge-
based

4

Hybrid/
Ensemble-based

Collaborative Filtering Models

Here the focus is on **collaborative filtering** techniques.

Collaborative filtering uses the collaborative power of other users' ratings to make recommendations. For example:

“You liked Movie A. Other people who liked Movie A also liked Movie B. Therefore, we recommend Movie B to you.”

Collaborative Filtering Models

ITEM-BASED COLLABORATIVE FILTERING

You liked this movie, which is rated by users very similarly to this other movie, so therefore you would like it.

USER-BASED COLLABORATIVE FILTERING

This other user rated movies very similarly to you. You would like this other movie they rated highly.



Item-Based Collaborative Filtering

ITEM-BASED COLLABORATIVE FILTERING

You liked this movie, which is rated by users very similarly to this other movie, so therefore you would like it.

We're doing this today!

USER-BASED COLLABORATIVE FILTERING

This other user rated movies very similarly to you. You would like this other movie they rated highly.

Explicit Rating Systems

Many rating systems are **explicit**, as in, you're rating the item on purpose. For example, many are **interval-based**:

Leave a review for this book:

Hated it!	Disliked it.	It was ok.	Liked it!	Loved it!
★☆☆☆☆	★★☆☆☆	★★★★☆	★★★★★	★★★★★

Explicit Rating Systems

Many rating systems are **explicit**, as in, you're rating the item on purpose. For example, many are **interval-based**:

Leave a review for this book:

Hated it!	Disliked it.	It was ok.	Liked it!	Loved it!
-2	-1	0	1	2

Explicit Rating Systems

Explicit ratings can be binary.

Did you like this book?

Dislike	Like
	
0	1

Implicit Rating Systems

Very often, ratings can be **unary**, in which case we might consider them **implicit**.

Did you like this book?

No feedback	Like
-	
-	1

Content-Based Recommender Systems

- In **content-based** systems, it's the actual descriptive attributes of the item itself that are used to make recommendations.
- Consider this example taken from *Recommender Systems*, by Charu C. Aggarwal:

Consider a situation where John has rated the movie Terminator highly, but we do not have access to the ratings of other users. Therefore, collaborative filtering methods are ruled out. However, the item description of Terminator contains similar genre keywords as other science fiction movies, such as Alien and Predator. In such cases, these movies can be recommended to John.

Content-Based Recommender Systems

- ☐ Using the skills you have right now, do you think you could make some kind of **content-based recommender system**?

Content-Based Recommender Systems

- Using the skills you have right now, do you think you could make some kind of **content-based recommender system**?

Sure! Probably! You could take the item's information and **cluster** it. Items in the same cluster would be more similar to each other and thus potential recommendations.

Even if you have text-data, you could use an NLP clustering technique such as **Latent Dirichlet Allocation (LDA)** which we don't learn about in our class, but is a popular technique for student capstones!

Knowledge-Based Recommender Systems

Knowledge-based systems use user-specifications to retrieve the best recommendation. A clear example of this is a website like **Yelp**:

I want food that...

- ☐ Offers dine-in
- ☐ Medium-price (\$\$)
- ☐ Good for kids
- ☐ Within 10 miles of my home
- ☐ Serves alcohol



Knowledge-Based Recommender Systems

Knowledge-based models are often used when:

1

The item being recommended is purchased by people rarely (cars, houses, etc.).

2

You don't have any starting data with user ratings (the "cold start" problem).

Knowledge-Based Recommender System

- How do you think you might carry out a **knowledge-based** recommender system?

Not everything needs to be complicated...

SELECT

restaurant_name,
restaurant_address

FROM

restaurants

WHERE

offers_dine_in
AND cost = '\$\$'
AND good_for_kids
AND distance <= 10
AND serves_alcohol



@brohrer@recsys.social
@_brohrer_

...

ML strategy tip

When you have a problem, build two solutions - a deep Bayesian transformer running on multicloud Kubernetes and a SQL query built on a stack of egregiously oversimplifying assumptions. Put one on your resume, the other in production. Everyone goes home happy.

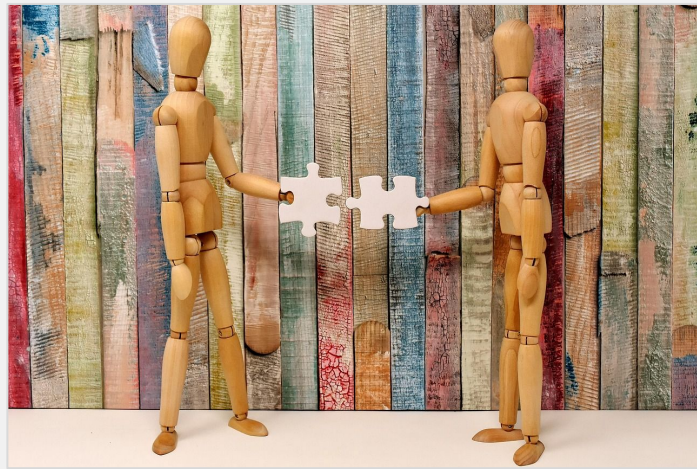
6:45 AM · Aug 12, 2021 · Twitter for iPhone



Hybrid or Ensemble-Based Systems

Few recommenders only rely on one system. It's common to combine them.

For example, our quintessential knowledge-based recommender **Yelp** also uses user reviews to create a collaborative filtering system.



Collaborative Filtering Example

Scenario

In this section you will look at collaborative filtering the MovieLens data set with cosine similarity.



The MovieLens Data

Movie	User	Rating
Movie 1	User 1	5
Movie 1	User 2	4
Movie 1	User 3	1
Movie 2	User 1	1
Movie 2	User 2	1
Movie 2	User 3	5
Movie 3	User 1	4
Movie 3	User 2	5
Movie 3	User 3	1

The MovieLens Data

Movie	User	Rating
Movie 1	User 1	5
Movie 1	User 2	4
Movie 1	User 3	1
Movie 2	User 1	1
Movie 2	User 2	1
Movie 2	User 3	5
Movie 3	User 1	4
Movie 3	User 2	5
Movie 3	User 3	1

PIVOT!

PIVOT!

	Movie 1	Movie 2	Movie 3
User 1	5	1	4
User 2	4	1	5
User 3	1	5	1

	User 1	User 2	User 3
Movie 1	5	4	1
Movie 2	1	1	5
Movie 3	4	5	1

Collaborative Filtering: **User-Based**

Remember in data science, we prefer

1 row = 1 observation

So in this view, we're considering **users** to be the object of study.

	Movie 1	Movie 2	Movie 3
User 1	5	1	4
User 2	4	1	5
User 3	1	5	1

Collaborative Filtering: **User-Based**

Remember in data science, we prefer

1 row = 1 observation

So in this view, we're considering **users** to be the object of study.

Who is most similar to me? I'm User 1

	Movie 1	Movie 2	Movie 3
User 1	5	1	4
User 2	4	1	5
User 3	1	5	1

Collaborative Filtering: Item-Based

Remember in data science, we prefer

1 row = 1 observation

So in this view, we're considering **movies** to be the object of study.

	User 1	User 2	User 3
Movie 1	5	4	1
Movie 2	1	1	5
Movie 3	4	5	1

Collaborative Filtering: Item-Based

Remember in data science, we prefer

1 row = 1 observation

So in this view, we're considering **movies** to be the object of study.

If I like Movie 1, what other movie might I like?

	User 1	User 2	User 3
Movie 1	5	4	1
Movie 2	1	1	5
Movie 3	4	5	1

Similarity Metric of Vectors

So, creating a movie recommender is as simple as measuring the **similarity of every movie based on its ratings!**

That is, the “similarity” of the movie vectors.



Similarity Metric of Vectors

- We'll learn something new for this, but any ideas based on stuff we already know how to do?

Similarity Example

	<i>LotR: The Fellowship of the Ring</i>	<i>Frozen</i>
Tim	4	1
Jamie	5	2
Heather	1	5

Similarity Example

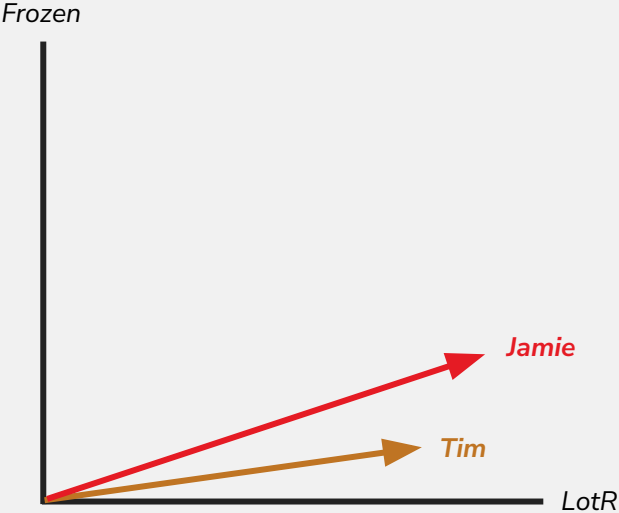
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Heather	1	5

Frozen

LotR

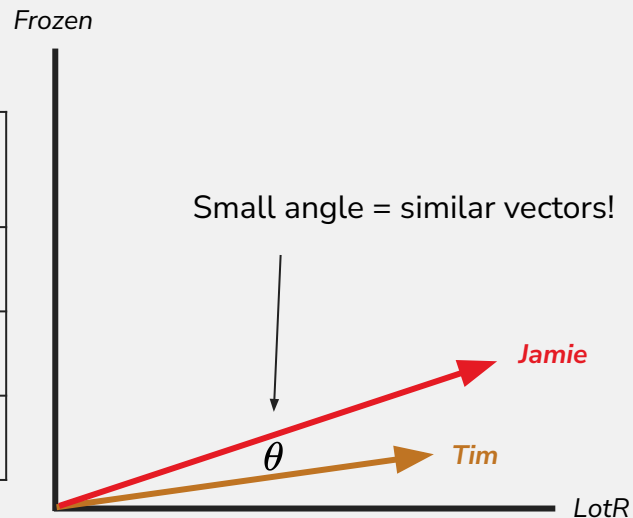
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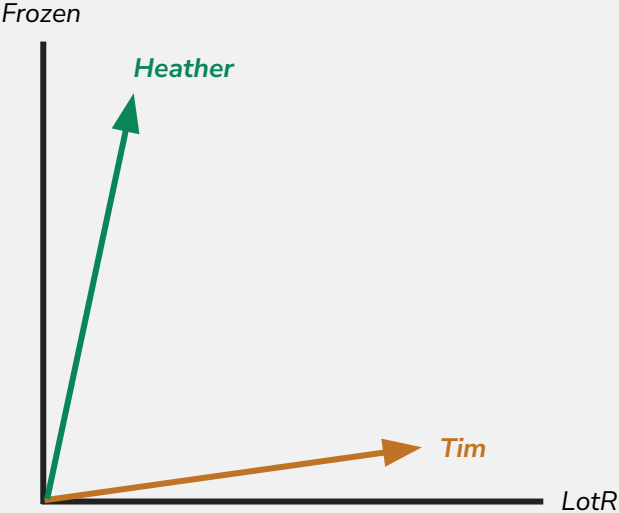
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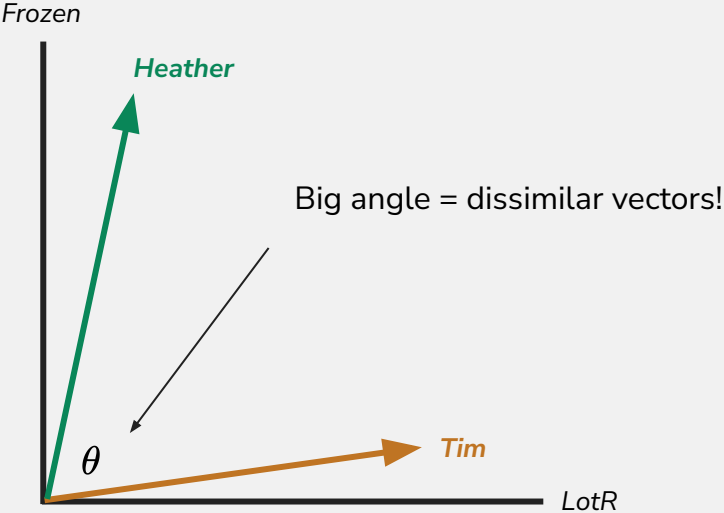
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Similarity Example

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Cosine Similarity

$$\text{cosine sim} = \cos \theta = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

Trig reminder:

- $\cos 0^\circ = 1$
- $\cos 90^\circ = 0$

Tim vs Jamie Example:

	x	y
t	4	1
j	5	2
h	1	5

$$\begin{aligned}\text{similarity} &= \frac{\mathbf{t} \cdot \mathbf{j}}{\|\mathbf{t}\| \|\mathbf{j}\|} \\&= \frac{4 \times 5 + 1 \times 2}{\sqrt{4^2 + 1^2} \sqrt{5^2 + 2^2}} \\&= \frac{22}{\sqrt{17} \sqrt{29}} \\&= 0.991\end{aligned}$$

Tim vs Heather Example:

	x	y
t	4	1
j	5	2
h	1	5

$$\begin{aligned}\text{similarity} &= \frac{\mathbf{t} \cdot \mathbf{h}}{\|\mathbf{t}\| \|\mathbf{h}\|} \\&= \frac{4 \times 1 + 1 \times 5}{\sqrt{4^2 + 1^2} \sqrt{1^2 + 5^2}} \\&= \frac{9}{\sqrt{17} \sqrt{26}} \\&= 0.428\end{aligned}$$

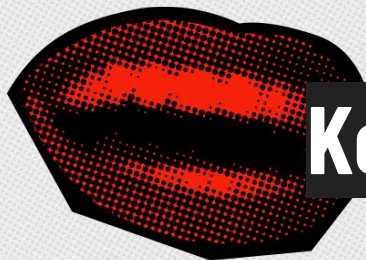
Collaborative Filtering

Let's see it for ourselves!

- Open **02-recommender-systems.ipynb**



Bring It Home



Key takeaways



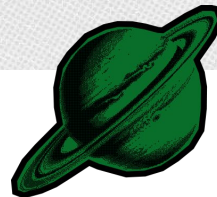
Recommender systems are useful

We use them every day for products and services online.



There are many ways to build recommendations

Collaborative filtering and Content-based are the most common.



Recommendations are based on similarity

Cosine similarity is a common metric.

